MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, ANALOG, ANALOG, KEY CODE ZSW

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic analog switch module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.10 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZSW.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/183-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of ±15.0 Vdc ±10% when driving the specified loads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

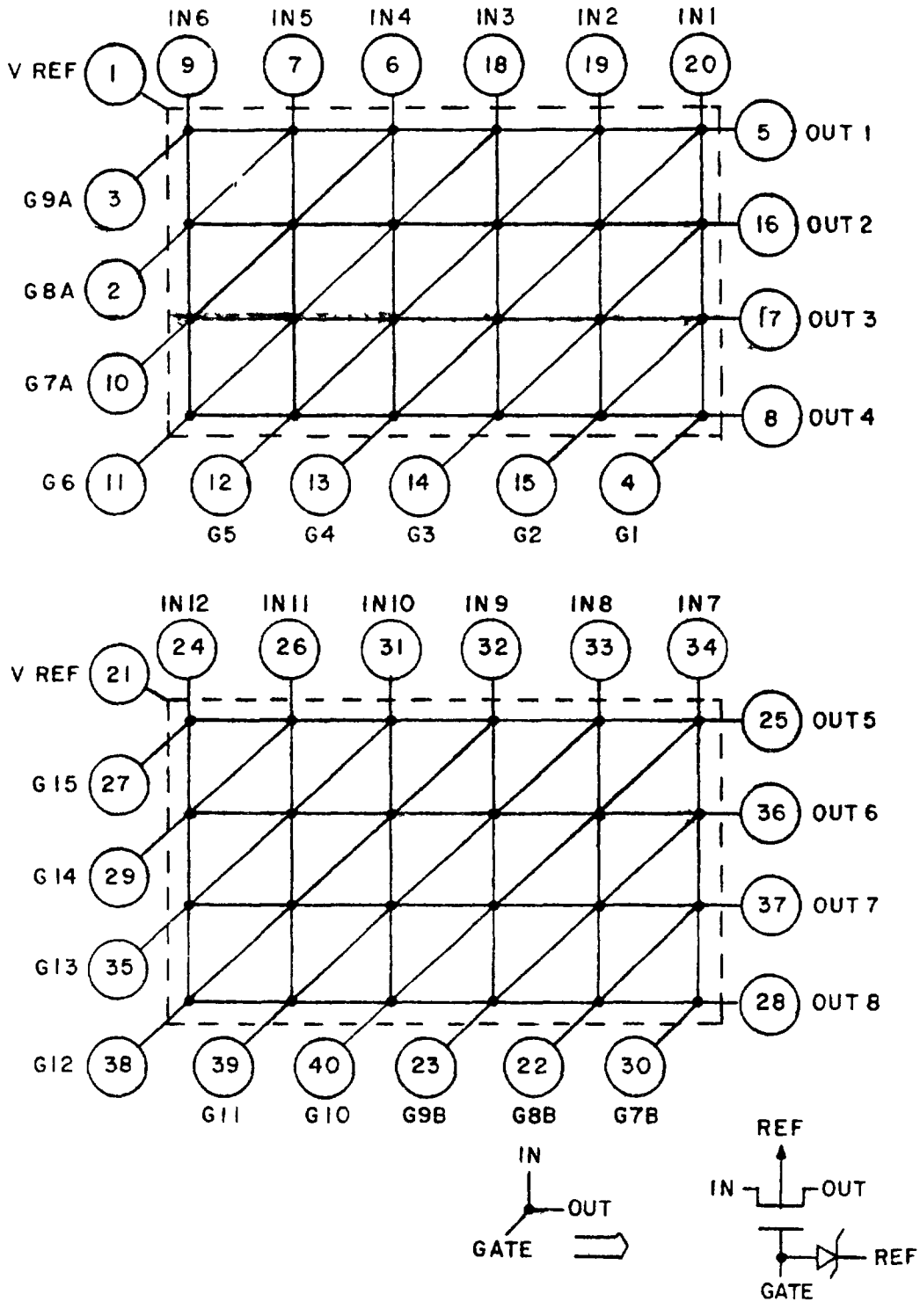
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES
1. IN = Input (source)
2 OUT = Output (drain)
3 G = Gate
4 V REF = Reference or body voltage.

FIGURE 1. ZSW module function diagram, analog switch.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A48	Stress (V_B - V_G = 35 V)	III	S	11.5	25.0	11.5	25.0	11.5	25.0	11.5	25.0	V
B	Isolation	IV	S	100	∞	100	∞	10	∞	10	∞	MΩ
C1-C48	ON resistance (100Ω) check	V	S	0	10	0	10	0.0	12.5	0.0	12.5	mV
D1-D48	ON resistance (200Ω) check	VI	S	0	20	0	20	0	25	0	25	mV
E1-E48	ON resistance (450Ω) check	VII	X	0	45	0	45	0	60	0	60	mV
F1-F12	Input leakage current	VIII	X	0.0	0.4	0.0	0.4	0.0	4.0	0.0	4.0	μA
G1-G8	Output leakage current	IX	X	0.0	0.4	0.0	0.4	0.0	5.0	0.0	5.0	μA
H1-H18	Gate leakage current	X	X	0.0	0.4	0.0	0.4	0.0	4.0	0.0	4.0	μA
J1-J48	Capacitance, gate to input	XI	Q	1	9	1	9	1	9	1	9	pF
K1-K48	Capacitance, gate to output	XI	Q	1	9	1	9	1	9	1	9	pF
L1-L48	Capacitance, channel	XI	Q	0	3	0	3	0	3	0	3	pF
M1-M12	Capacitance, input to reference	XII	Q	5	30	5	30	5	30	5	30	pF
N1-N8	Capacitance, output to reference	XII	Q	5	25	5	25	5	25	5	25	pF
P1-P48	Threshold voltage (gate to input) check	XIII	X	0	200	0	200	0	200	0	200	mV
Q1-Q56	Output isolation	XIV	X	0.0	1.7	0.0	1.7	0.0	1.7	0.0	1.7	mVrms
R1-R12	Output distortion	XV	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	%
S1-S48	Output switching time (high level), negative 5 V output	XVI	Q	0	6,000	0	6,000	0	6,000	0	6,000	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T1-T48	Output switching time (low level), negative 5 V output	XVII	Q	0	200	0	200	0	200	0	200	ns
U1-U48	Output switching time (high level), 0 V output	XVIII	S	0	5,000	0	5,000	0	5,000	0	5,000	ns
V1-V48	Output switching time (low level), 0 V output	XIX	S	0	200	0	200	0	200	0	200	ns
W1-W48	Output switching time (high level), positive 5 V output	XX	Q	0	2,500	0	2,500	0	2,500	0	2,500	ns
X1-X48	Output switching time (low level), positive 5 V output	XXI	Q	0	200	0	200	0	200	0	200	ns
Y1-Y48	Output switching time, most positive peak, 0 V output	XXII	S	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	V
Z1-Z48	Output switching time, most negative peak, 0 V output	XXIII	S	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2 5	V
AA1-AA48	Output switching time, most positive peak, +5 V output	XXIV	Q	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	V
BB1-BB48	Output switching time, most negative peak, +5 V output	XXV	Q	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through H and P through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through BB shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/183 supersedes Department of the Navy, Naval Ordance Systems Command Drawing No. 10001-2658686. Part number M28787/183-1 supersedes part numbers 2658686 and M28787/183.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits of table I.

Power dissipation	1.75 W per module 500 mW per output maximum
Maximum voltage ratings	
V_B-V_S	-0.3 to 35 V
V_B-V_D	-0.3 to 35 V
V_B-V_S	±30 V
V_B-V_G	35 V

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Maximum current rating I_S, I_D		50 mA
Gate threshold		-6.0 V gate to source
ON resistance	Initial	End-of-life
V_{BD} = 0, V_{GD} = -30 V, I_S = 100 µA	100Ω	125Ω maximum
V_{DB} = 10 V, V_{GD} = -20 V, I_S = 100 µA	200Ω	250Ω maximum
V_{DB} = 20 V, V_{GD} = -10 V, I_S = 100 µA	450Ω	600Ω maximum
Gate leakage current		4 µA minimum
Input leakage current		4 µA maximum
Output leakage current		5 µA maximum
Gate to input capacitance		9 pF maximum
Gate to output capacitance		9 pF maximum
Channel capactance		3 pF maximum
Output distortion (V_{IN} = 3.57 Vrms)		0.5 percent maximum
Output attenuation per isolation (R_L = 1 kΩ, frequency = 20 kHz)		65 dB minimum

6.5 Module function. This module is a dual expandable 4 lines out of 6 analog switching matrix, consisting of 24 analog switching points.

6.5.1 Module operation. The module operates as shown in table II with all gates off except the gate indicated in the table.

TABLE II. Module operation relationship.

Gate on (only)	Outputs			
	Out 1	Out 2	Out 3	Out 4
G7	IN4	IN5	IN6	OFF
G6	IN3	IN4	IN5	IN6
G5	IN2	IN3	IN4	IN5
G4	IN1	IN2	IN3	IN4
G3	OFF	IN1	IN2	IN3
G2	OFF	OFF	IN1	IN2
G1	OFF	OFF	OFF	IN1

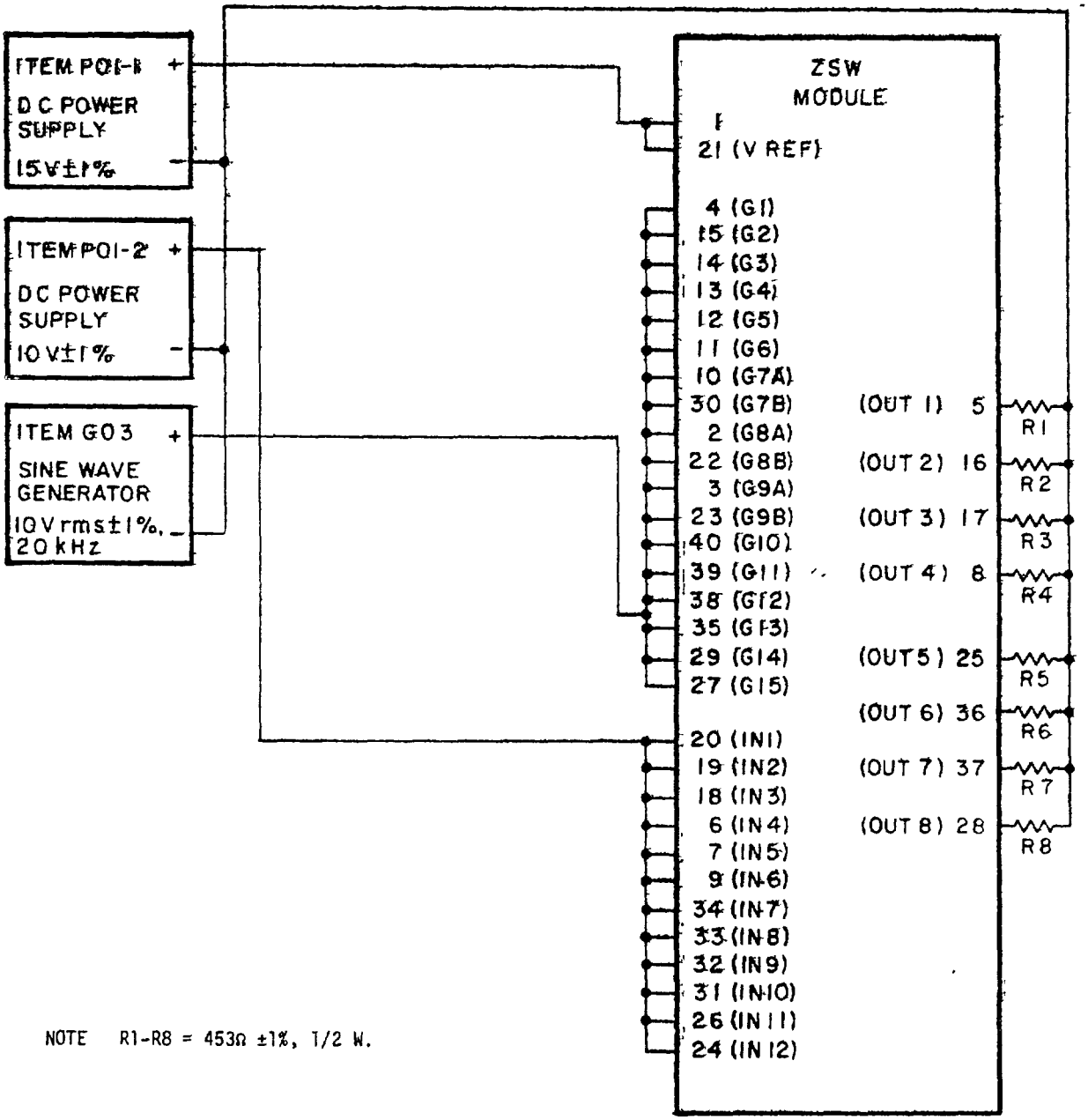
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
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NOTE R1-R8 = 453Ω ±1%, 1/2 W.

FIGURE 2 Operating temperature, component temperature, and life test circuit

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TABLE III. Stress test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A1	Stress (V_B - V_G = 35 V)	3	9	5	11.5	25.0	V	3
A2		2	7	5				
A3		2	9	16				
A4		10	6	5				
A5		10	7	16				
A6		10	9	17				
A7		11	18	5				
A8		11	6	16				
A9		11	7	17				
A10		11	9	8				
A11		12	19	5				
A12		12	18	16				
A13		12	6	17				
A14		12	7	8				
A15		13	20	5				
A16		13	19	16				
A17		13	18	17				

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TABLE III. Stress test - Continued.

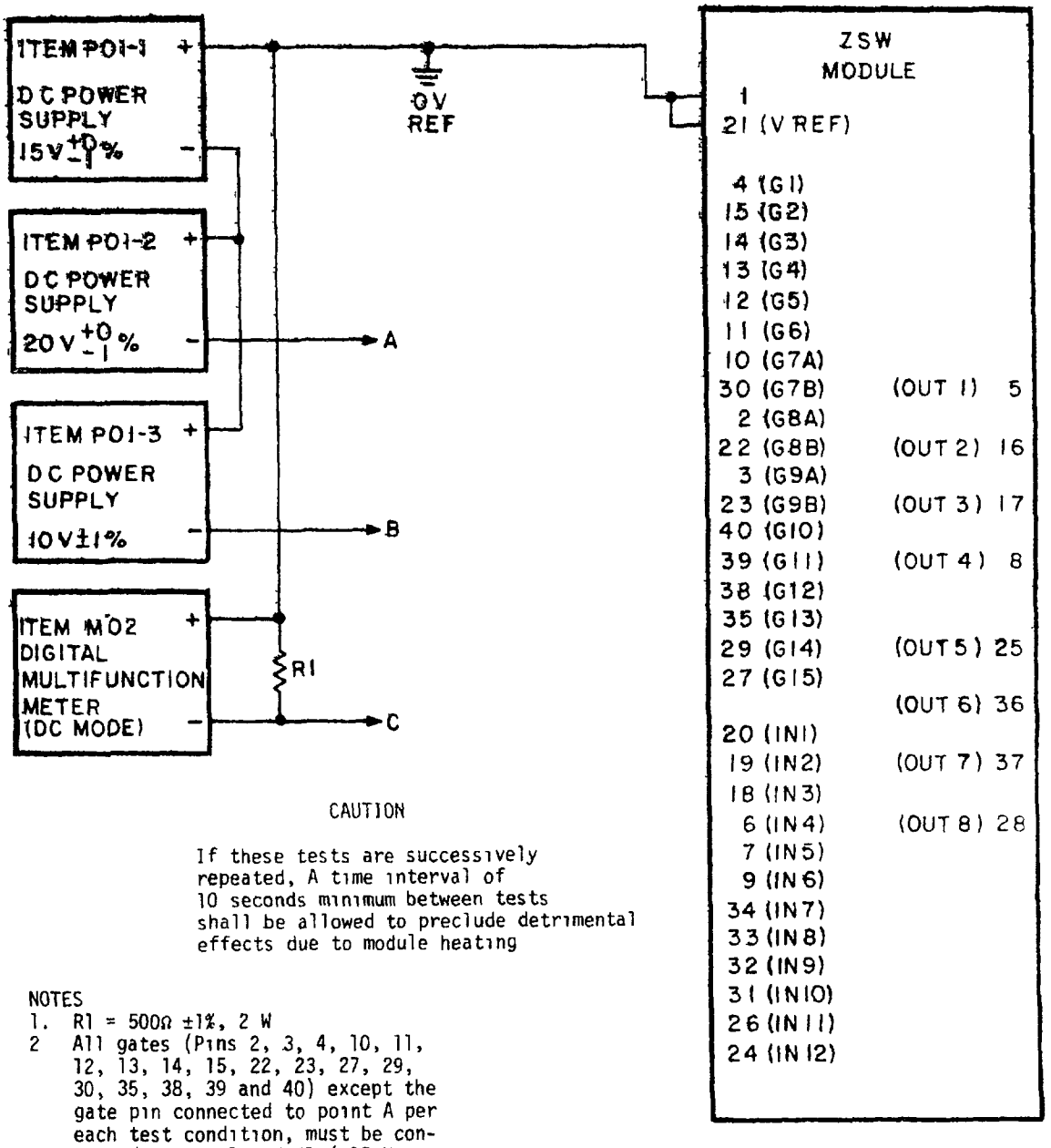
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A18	Stress (V_B - V_G = 35 V)	13	6	8	11.5	25.0	V	3
A19		14	20	16				
A20		14	19	17				
A21		14	18	8				
A22		15	20	17				
A23		15	19	8				
A24		4	20	8				
A25		27	24	25				
A26		29	26	25				
A27		29	24	36				
A28		35	31	25				
A29		35	26	36				
A30		35	24	37				
A31		38	32	25				
A32		38	31	36				
A33		38	26	37				
A34		38	24	28				
A35		39	33	25				
A36		39	32	36				
A37		39	31	37				
A38		39	26	28				

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TABLE III. Stress test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A39	Stress (V_B - V_G = 35 V)	40	34	25	11.5	25.0	V	3
A40	↓	40	33	36	↓	↓	↓	↓
A41		40	32	37				
A42		40	31	28				
A43		23	34	36				
A44		23	33	37				
A45		23	32	28				
A46		22	34	37				
A47		22	33	28				
A48		30	34	28				

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CAUTION

If these tests are successively repeated, A time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating

NOTES
1. R1 = 500Ω ±1%, 2 W
2 All gates (Pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point A per each test condition, must be connected to pin 1 and 21 (+15 V turned off)

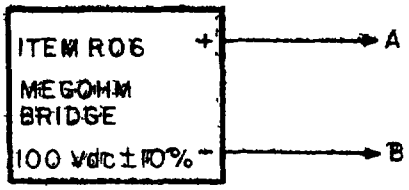
FIGURE 3 Stress test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Isolation	All pins	Frame	100	∞	MΩ	4

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

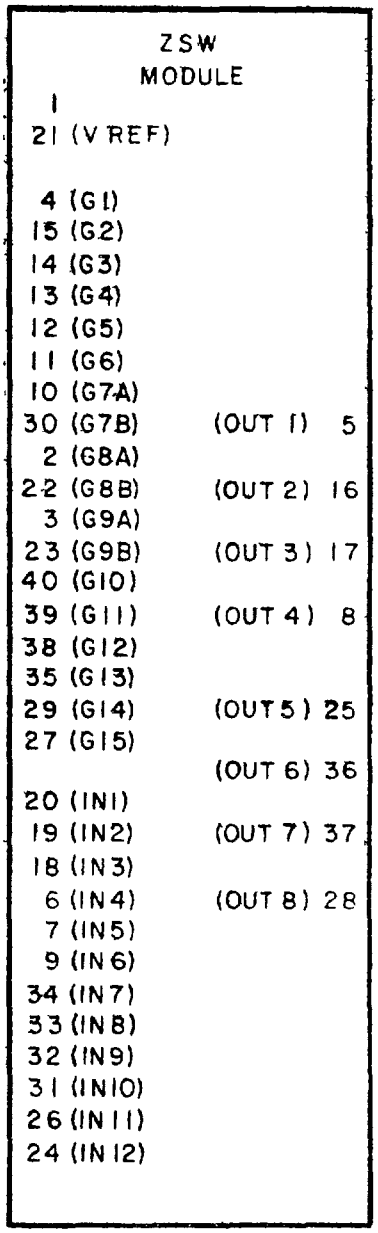


FIGURE 4. Isolation test circuit for table IV

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TABLE V ON resistance (100Ω) check test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	ON resistance (100Ω) check (V_{BD} = 0, V_{GD} = -30 V, I_S = 100 μA)	3	9	5	0	10	mV	5
C2		2	7	5				
C3		2	9	16				
C4		10	6	5				
C5		10	7	16				
C6		10	9	17				
C7		11	18	5				
C8			6	16				
C9			7	17				
C10			9	8				
C11		12	19	5				
C12			18	16				
C13			6	17				
C14			7	8				
C15		13	20	5				
C16			19	16				
C17			18	17				
C18			6	8				
C19		14	20	16				

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TABLE V. ON resistance (100Ω) check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C20	ON resistance (100Ω) check (V_{BD} = 0, V_{GD} = -30 V, I_S = 100 μA)	14	19	17	0	10	mV	5
C21		14	18	8				
C22		15	20	17				
C23		15	19	8				
C24		4	20	8				
C25		27	24	25				
C26		29	26	25				
C27		29	24	36				
C28		35	31	25				
C29		35	26	36				
C30		35	24	37				
C31		38	32	25				
C32		38	31	36				
C33		38	26	37				
C34		38	24	28				
C35		39	33	25				
C36		39	32	36				
C37		39	31	37				
C38		39	26	28				

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TABLE V. ON resistance (100%) check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C39	ON resistance (100%) check (V_{DB} = 0, V_{GD} = -30 V, I_S = 100 µA)	40	34	25	0	10	mV	5
C40	↓	40	33	36	↓	↓	↓	↓
C41		40	32	37				
C42		40	31	28				
C43		23	34	36				
C44		23	33	37				
C45		23	32	28				
C46		22	34	37				
C47		22	33	28				
C48	↓	30	34	28	↓	↓	↓	↓

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TABLE VI. ON resistance (200Ω) check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	ON resistance (200Ω) check (V_{DB} = -10 V, V_{GD} = -20 V, I_S = 100 µA)	3	9	5	0	20	mV	5
D2		2	7	5				
D3		2	9	16				
D4		10	6	5				
D5		10	7	16				
D6		10	9	17				
D7		11	18	5				
D8			6	16				
D9			7	17				
D10			9	8				
D11		12	19	5				
D12			18	16				
D13			6	17				
D14			7	8				
D15		13	20	5				
D16			19	16				
D17			18	17				
D18			6	8				
D19		14	20	15				

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TABLE VI. ON resistance (200Ω) check test - Continued.

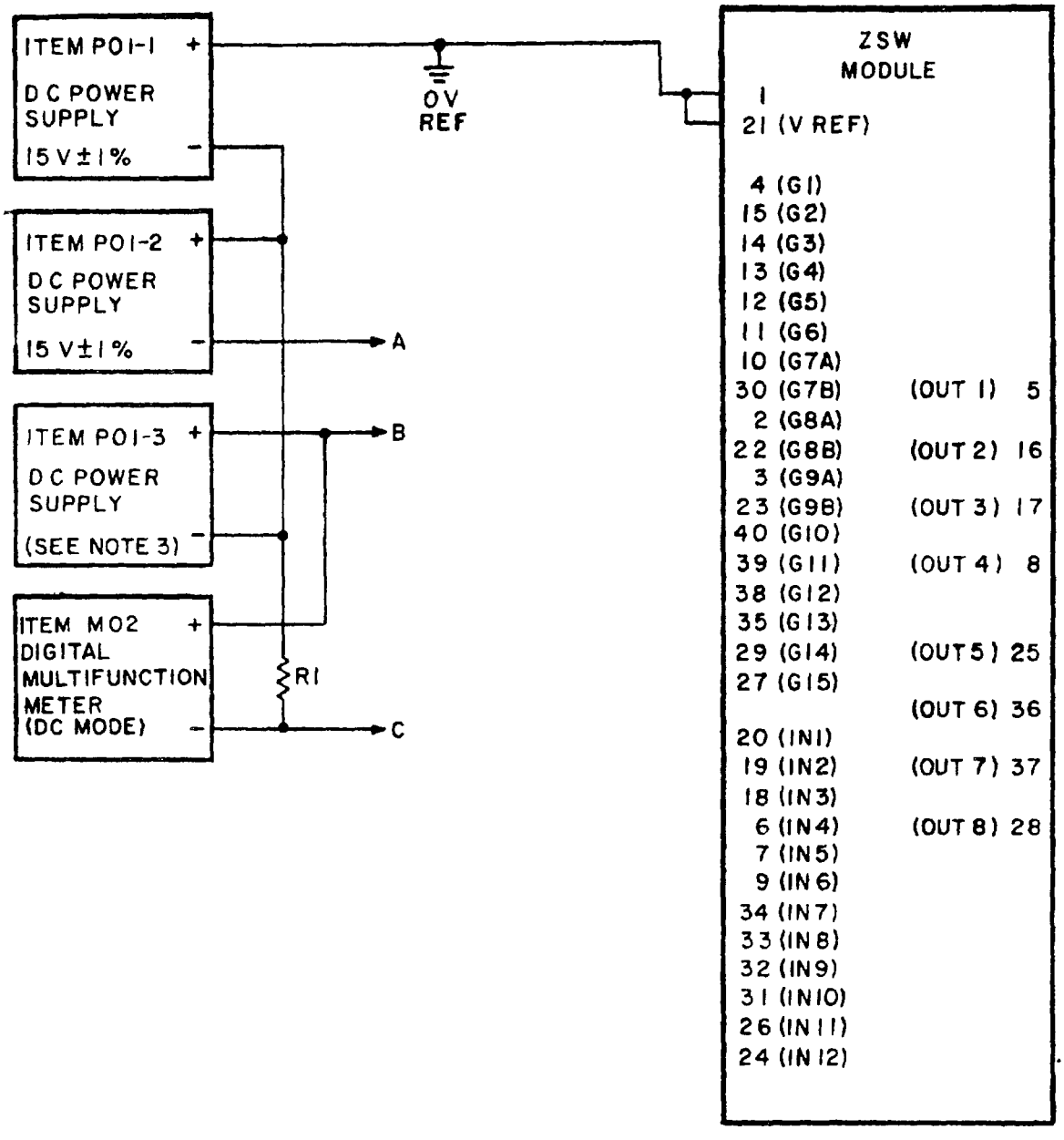
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D20	ON resistance (200Ω) check (V_{DB} = -10 V, V_{GD} = -20 V, I_S = 100 µA)	14	19	17	0	20	mV	5
D21	↓	14	18	8	↓	↓	↓	↓
D22		15	20	17				
D23		15	19	8				
D24		4	20	8				
D25		27	24	25				
D26		29	26	25				
D27		29	24	36				
D28		35	31	25				
D29		35	26	36				
D30		35	24	37				
D31		38	32	25				
D32		↓	31	36				
D33			26	37				
D34		↓	24	28				
D35		39	33	25				
D36		↓	32	36				
D37			31	37				
D38	↓	↓	26	28	↓	↓	↓	↓

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TABLE VI. ON resistance (200Ω) check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D39	ON resistance (200Ω) check (V_{DB} = -10 V, V_{GD} = -20 V, I_S = 100 µA)	40	34	25	0	20	mV	5
D40			33	36				
D41			32	37				
D42			31	28				
D43		23	34	36				
D44		23	33	37				
D45		23	32	28				
D46		22	34	37				
D47		22	33	28				
D48		30	34	28				

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NOTES
1. R1 = 49.9 kΩ ±1%, 1/8 W for test conditions D1-D48.
   = 150 kΩ ±1%, 1/8 W for test conditions C1-C48.
2. All gates (pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point A per each test condition, must be connected to pin 1 and 21 (+15 V turned off).
3. Item PO1-3 = 15 V ±1% for test conditions C1-C48 and 5.0 V ±1% for test conditions D1-D48

FIGURE 5. ON resistance (100 and 200Ω) check test circuit for tables V and VI.

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TABLE VII. ON resistance (450Ω) check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	ON resistance (450Ω) check (V_{DB} = -20 V, V_{GB} = -10 V, I_S = 100 µA)	3	9	5	0	45	mV	6
E2		2	7	5				
E3		2	9	16				
E4		10	6	5				
E5		10	7	16				
E6		10	9	17				
E7		11	18	5				
E8			6	16				
E9			7	17				
E10			9	8				
E11		12	19	5				
E12			18	16				
E13			6	17				
E14			7	8				
E15		13	20	5				
E16			19	16				
E17			18	17				
E18			6	8				
E19		14	20	16				

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TABLE VII ON resistance (450Ω) check test - Continued.

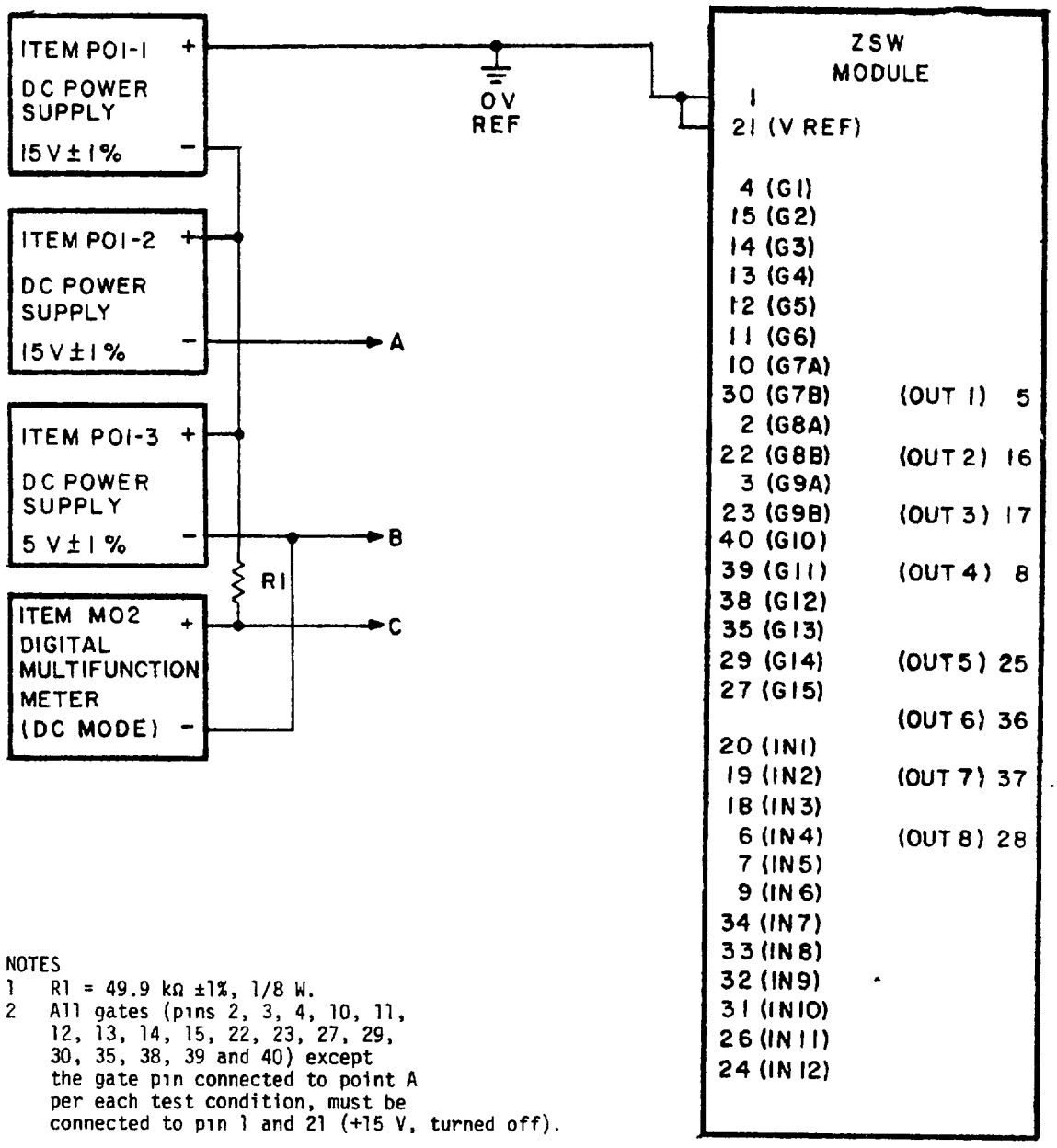
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E20	ON resistance (450Ω) check (V_{DB} = -20 V, V_{GD} = -10 V, I_S = 100 µA)	14	19	17	0	45	mV	6
E21		14	18	8				
E22		15	20	17				
E23		15	19	8				
E24		4	20	8				
E25		27	24	25				
E26		29	26	25				
E27		29	24	36				
E28		35	31	25				
E29		35	26	36				
E30		35	24	37				
E31		38	32	25				
E32			31	36				
E33			26	37				
E34			24	28				
E35		39	33	25				
E36			32	36				
E37			31	37				
E38			26	28				

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TABLE VII. ON resistance (450Ω) check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E39	ON resistance (450Ω) check (V_{DB} = -20 V, V_{GD} = -10 V, I_S = 100 µA)	40	34	25	0	45	mV	6
E40	↓	40	33	36	↓	↓	↓	↓
E41		40	32	37				
E42		40	31	28				
E43		23	34	36				
E44		23	33	37				
E45		23	32	28				
E46		22	34	37				
E47		22	33	28				
E48	↓	30	34	28	↓	↓	↓	↓

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NOTES
1 R1 = 49.9 kΩ ±1%, 1/8 W.
2 All gates (pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point A per each test condition, must be connected to pin 1 and 21 (+15 V, turned off).

FIGURE 6 ON resistance (450Ω) check test circuit for table VII

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TABLE VIII. Input leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	IN 1 Input leakage current	20	0.0	0.4	µA	7
F2	IN 2	19	↓	↓	↓	↓
F3	IN 3	18				
F4	IN 4	6				
F5	IN 5	7				
F6	IN 6	9				
F7	IN 7	34				
F8	IN 8	33				
F9	IN 9	32				
F10	IN 10	31				
F11	IN 11	26				
F12	IN 12	24				

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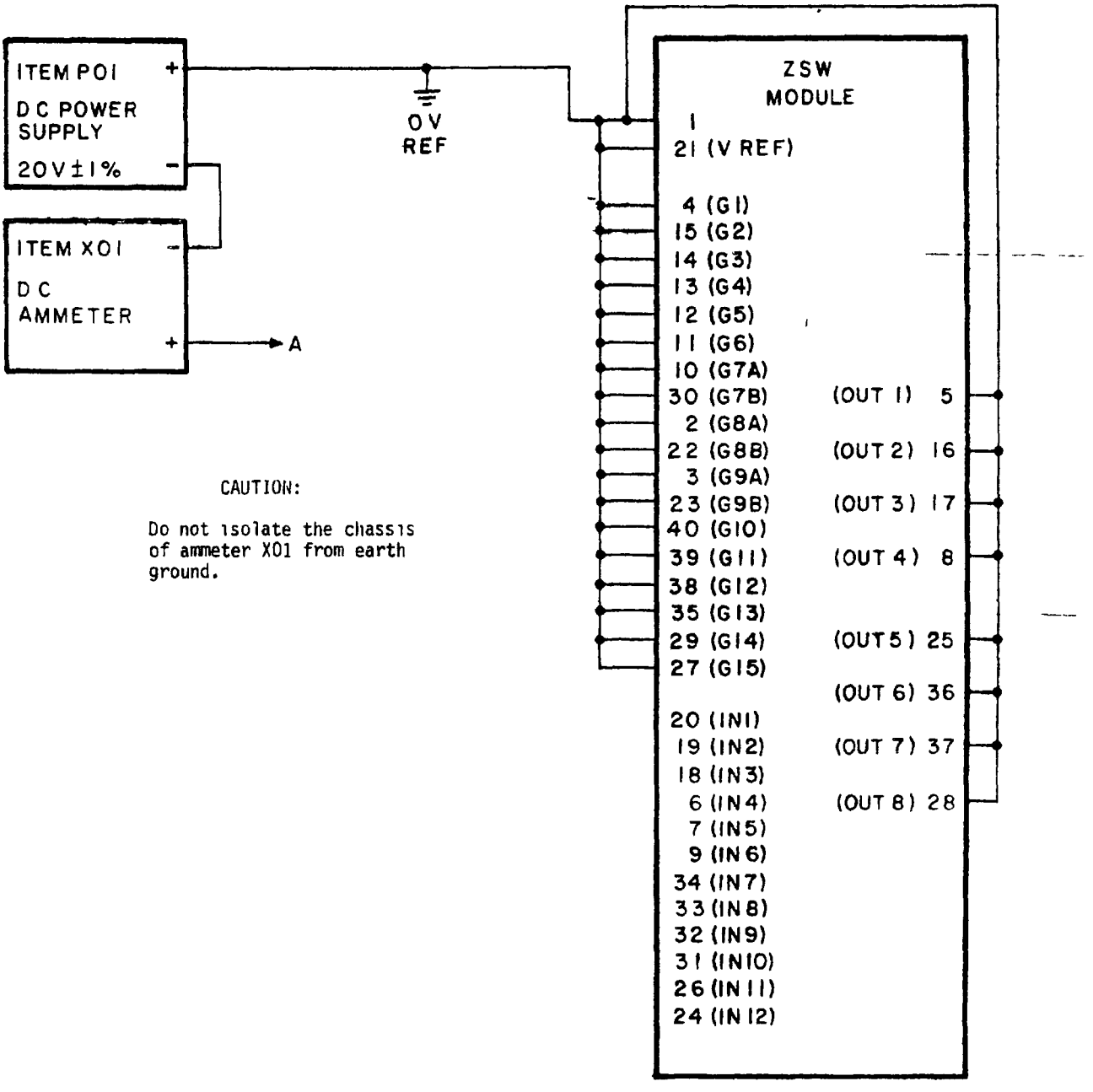


FIGURE 7. Input leakage current test circuit for table VIII.

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TABLE IX. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	OUT 1 Output leakage current	5	0.0	0.4	µA	8
G2	OUT 2	16				
G3	OUT 3	17				
G4	OUT 4	8				
G5	OUT 5	25				
G6	OUT 6	36				
G7	OUT 7	37				
G8	OUT 8	28				

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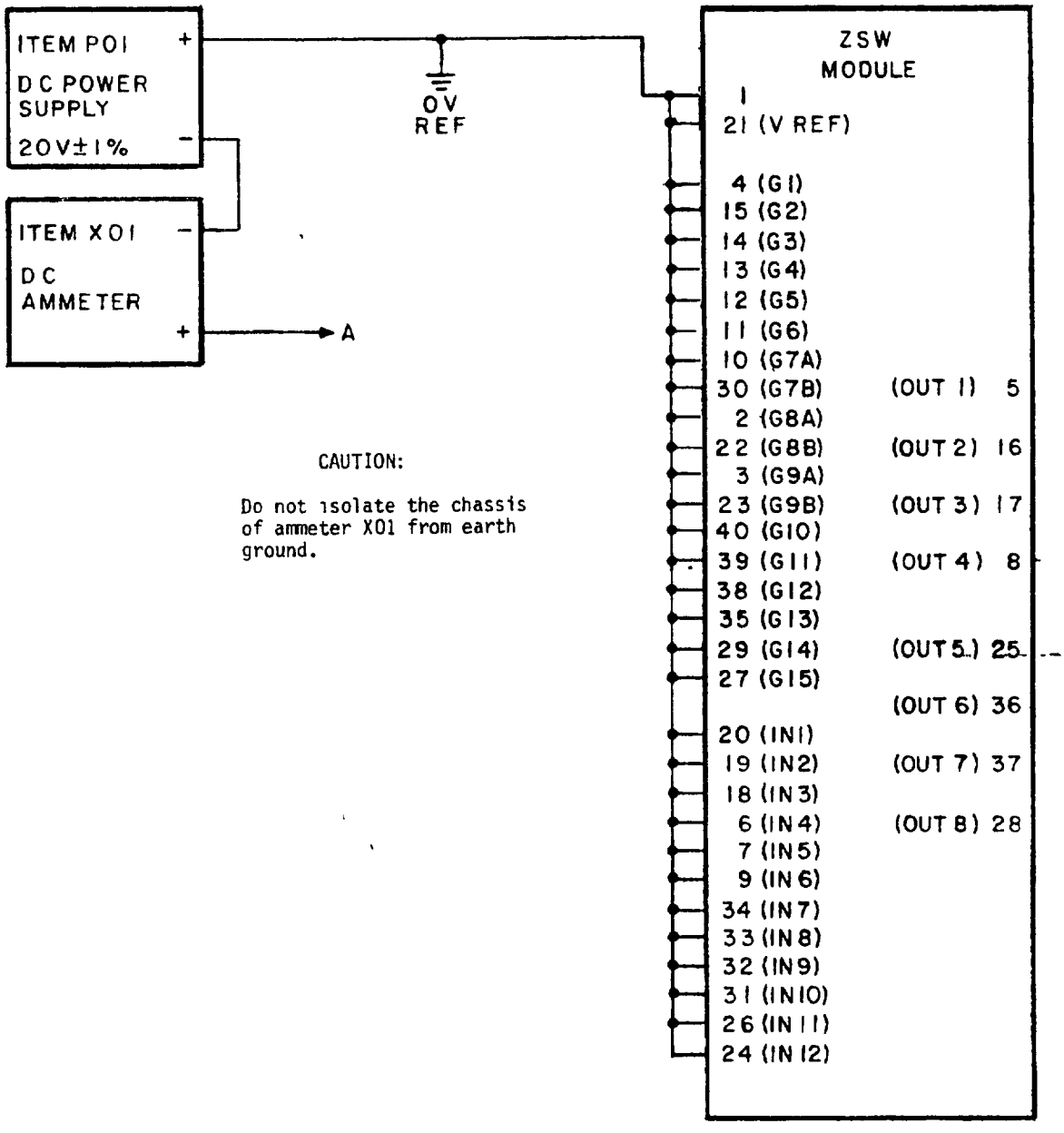


FIGURE 8. Output leakage current test circuit for table IX.

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TABLE X. Gate leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	G1 Gate leakage current	4	0.0	0.4	µA	9
H2	G7B	30	↓	↓	↓	↓
H3	G9A	3				
H4	G15	27				
H5	G2	15				
H6	G8A	2				
H7	G8B	22				
H8	G14	29				
H9	G3	14				
H10	G7A	10				
H11	G9B	23				
H12	G13	35				
H13	G4	13				
H14	G5	12				
H15	G6	11				
H16	G10	40				
H17	G11	39				
H18	G12	38				

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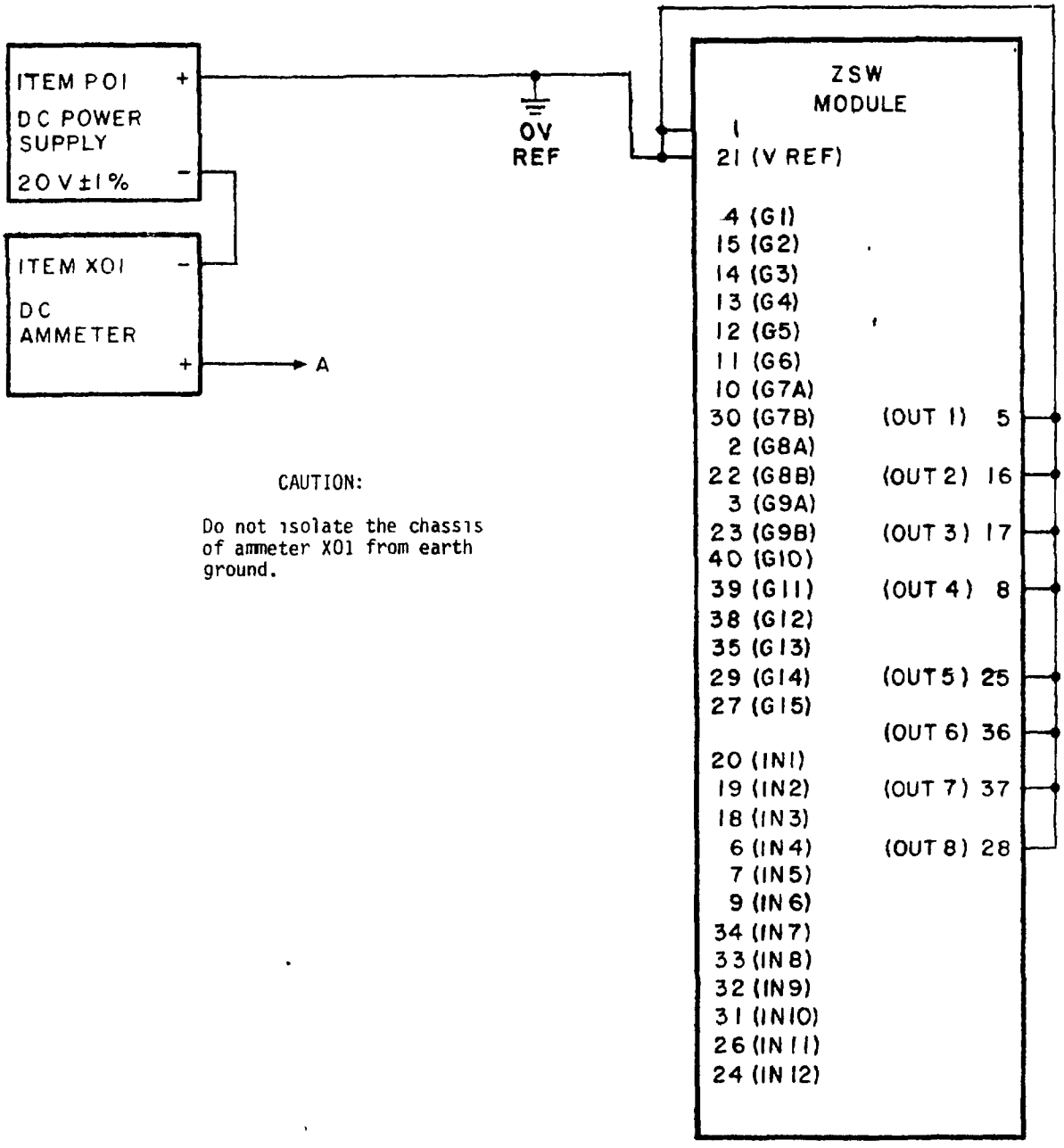


FIGURE 9. Gate leakage current test circuit for table X.

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TABLE XI Capacitance (gate to input, gate to output, and channel) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Capacitance, gate to input	3	9	5, 1, 21	1	9	pF	10
J2		2	7					
J3		10	6					
J4		11	18					
J5		12	19					
J6		13	20					
J7		2	9	16, 1, 21				
J8		10	7					
J9		11	6					
J10		12	18					
J11		13	19					
J12		14	20					
J13		10	9	17, 1, 21				
J14		11	7					
J15		12	6					
J16		13	18					
J17		14	19					
J18		15	20					
J19		11	9	8, 1, 21				
J20		12	7	8, 1, 21				
J21		13	6	8, 1, 21				

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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J22	Capacitance, gate to input	14	18	8, 1, 21	1	9	pF	10
J23		15	19	8, 1, 21				
J24		4	20	8, 1, 21				
J25		27	24	25, 1, 21				
J26		29	26					
J27		35	31					
J28		38	32					
J29		39	33					
J30		40	34					
J31		29	24	36, 1, 21				
J32		35	26					
J33		38	31					
J34		39	32					
J35		40	33					
J36		23	34					
J37		35	24	37, 1, 21				
J38		38	26					
J39		39	31					
J40		40	32					
J41		23	33					

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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J42	Capacitance, gate to input	22	34	37, 1, 21	1	9	pF	10
J43	↓	38	24	28, 1, 21	↓	↓	↓	↓
J44		39	26	↓				
J45		40	31					
J46		23	32					
J47		22	33					
J48		30	34					
K1	Capacitance, gate to output	13	5	20, 1, 21				
K2	↓	14	16	↓				
K3		15	17					
K4		4	8					
K5		12	5	19, 1, 21				
K6		13	16	↓				
K7		14	17					
K8		15	8					
K9		11	5	18, 1, 21				
K10		12	16	↓				
K11		13	17					
K12	↓	14	8		↓	↓	↓	↓

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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K13	Capacitance, gate to output	10	5	6, 1, 21	1	9	pF	10
K14		11	16					
K15		12	17					
K16		13	8					
K17		2	5	7, 1, 21				
K18		10	16					
K19		11	17					
K20		12	8					
K21		3	5	9, 1, 21				
K22		2	16					
K23		10	17					
K24		11	8					
K25		40	25	34, 1, 21				
K26		23	36					
K27		22	37					
K28		30	28					
K29		39	25	33, 1, 21				
K30		40	36					
K31		23	37					
K32		22	28					

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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K33	Capacitance, gate to output	38	25	32, 1, 21	1	9	pF	10
K34	↓	39	36	↓	↓	↓	↓	↓
K35	↓	40	37	↓	↓	↓	↓	↓
K36	↓	23	28	↓	↓	↓	↓	↓
K37	↓	35	25	31, 1, 21	↓	↓	↓	↓
K38	↓	38	36	↓	↓	↓	↓	↓
K39	↓	39	37	↓	↓	↓	↓	↓
K40	↓	40	28	↓	↓	↓	↓	↓
K41	↓	29	25	26, 1, 21	↓	↓	↓	↓
K42	↓	35	36	↓	↓	↓	↓	↓
K43	↓	38	37	↓	↓	↓	↓	↓
K44	↓	39	28	↓	↓	↓	↓	↓
K45	↓	27	25	24, 1, 21	↓	↓	↓	↓
K46	↓	29	36	↓	↓	↓	↓	↓
K47	↓	35	37	↓	↓	↓	↓	↓
K48	↓	38	28	↓	↓	↓	↓	↓
L1	Capacitance, channel	8	20	4, 1, 21	0	3	↓	↓
L2	↓	8	19	15, 1, 21	↓	↓	↓	↓
L3	↓	17	20	15, 1, 21	↓	↓	↓	↓
L4	↓	8	18	14, 1, 21	↓	↓	↓	↓

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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued.

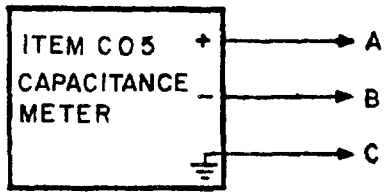
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L5	Capacitance, channel	17	19	14, 1, 21	0	3	pF	10
L6		16	20	14, 1, 21				
L7		8	6	13, 1, 21				
L8		17	18					
L9		16	19					
L10		5	20					
L11		8	7	12, 1, 21				
L12		17	6					
L13		16	18					
L14		5	19					
L15		8	9	11, 1, 21				
L16		17	7					
L17		16	6					
L18		5	18					
L19		17	9	10, 1, 21				
L20		16	7	10, 1, 21				
L21		5	6	10, 1, 21				
L22		16	9	2, 1, 21				
L23		5	7	2, 1, 21				

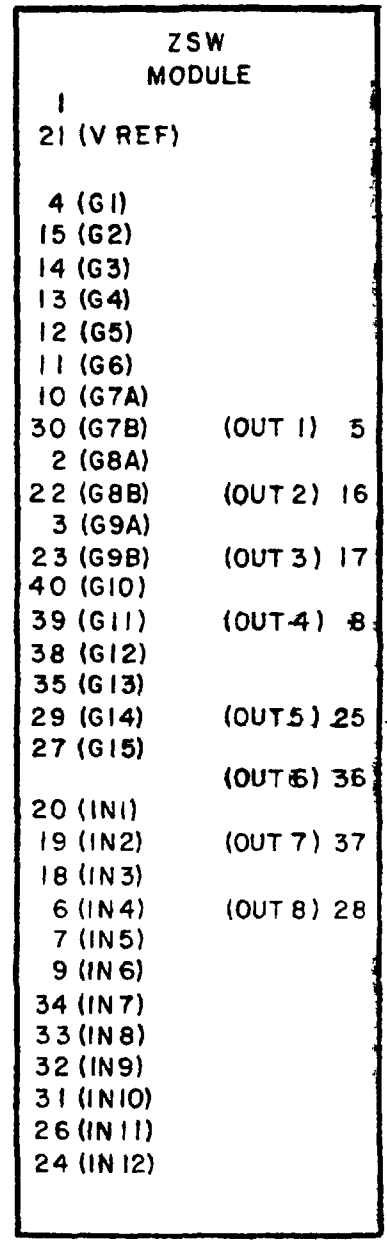
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TABLE XI. Capacitance (gate to input, gate to output, and channel) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L24	Capacitance, channel	5	9	3, 1, 21	0	3	pF	10
L25		28	34	30, 1, 21				
L26		28	33	22, 1, 21				
L27		37	34	22, 1, 21				
L28		28	32	23, 1, 21				
L29		37	33	23, 1, 21				
L30		36	34	23, 1, 21				
L31		28	31	40, 1, 21				
L32		37	32					
L33		36	33					
L34		25	34					
L35		28	26	39, 1, 21				
L36		37	31					
L37		36	32					
L38		25	33					
L39		28	24	38, 1, 21				
L40		37	26					
L41		36	31					
L42		25	32					
L43		37	24	35, 1, 21				
L44		36	26	35, 1, 21				
L45		25	31	35, 1, 21				
L46		36	26	29, 1, 21				
L47		25	24	29, 1, 21				
L48		25	24	27, 1, 21				

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NOTES
1. Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or leads to any of the module pins other than those specified. Stray capacitance added to the other pins may affect the reading

FIGURE 10 Capacitance (gate to input, gate to output, and channel) test circuit for table XI.

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TABLE XII. Capacitance (input to reference and output to reference) test.

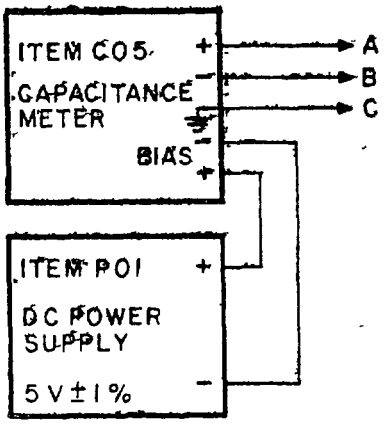
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Capacitance, input to reference	1	20	4, 15, 14, 13, 5, 16, 17, 8	5	30	pF	11
M2			19	15, 14, 13, 12, 5, 16, 17, 8				
M3			18	14, 13, 12, 11, 5, 16, 17, 8				
M4			6	13, 12, 11, 10, 5, 16, 17, 8				
M5			7	12, 11, 10, 2, 5, 16, 17, 8				
M6			9	11, 10, 2, 3, 5, 16, 17, 8				
M7		21	34	30, 22, 23, 40, 25, 36, 37, 28				
M8			33	22, 23, 40, 39, 25, 36, 37, 28				
M9			32	23, 40, 39, 38, 25, 36, 37, 28				
M10			31	40, 39, 38, 35, 25, 36, 37, 28				
M11			26	39, 38, 35, 29, 25, 36, 37, 28				

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TABLE XII. Capacitance (input to reference and output to reference) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M12	Capacitance, input to reference	21	24	38, 35, 29, 27, 25, 36, 37, 28	5	30	pF	11
N1	Capacitance, output to reference	1	5	9, 7, 6, 18, 19, 20, 3, 2, 10, 11, 12, 13		25		
N2			16	9, 7, 6, 18, 19, 20, 2, 10, 11, 12, 13, 14				
N3			17	9, 7, 6, 18, 19, 20, 10, 11, 12, 13, 14, 15				
N4			8	9, 7, 6, 18, 19, 20, 11, 12, 13, 14, 15, 4				
N5		21	25	24, 26, 31, 32, 33, 34, 27, 29, 35, 38, 39, 40				
N6			36	24, 26, 31, 32, 33, 34, 29, 35, 38, 39, 40, 23				
N7			37	24, 26, 31, 32, 33, 34, 35, 38, 39, 40, 23, 22				
N8			28	24, 26, 31, 32, 33, 34, 38, 39, 40, 23, 22, 30				

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ZSW MODULE		
1		
21 (V REF)		
4 (G1)		
15 (G2)		
14 (G3)		
13 (G4)		
12 (G5)		
11 (G6)		
10 (G7A)		
30 (G7B)	(OUT 1)	5
2 (G8A)		
22 (G8B)	(OUT 2)	16
3 (G9A)		
23 (G9B)	(OUT 3)	17
40 (G10)		
39 (G11)	(OUT 4)	8
38 (G12)		
35 (G13)		
29 (G14)	(OUT5)	25
27 (G15)		
	(OUT 6)	36
20 (IN1)		
19 (IN2)	(OUT 7)	37
18 (IN3)		
6 (IN4)	(OUT 8)	28
7 (IN5)		
9 (IN6)		
34 (IN7)		
33 (IN8)		
32 (IN9)		
31 (IN10)		
26 (IN11)		
24 (IN12)		

NOTES
1 Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or leads to any of the module pins other than those specified. Stray capacitance added to the other pins may affect the reading.

FIGURE 11 Capacitance (input to reference and output to reference) test circuit for table XII

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TABLE XIII. Threshold voltage (gate to input) check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Threshold voltage, gate to input, check	20	4	8	0	200	mV	12
P2			15	17				
P3			14	16				
P4			13	5				
P5		19	15	8				
P6			14	17				
P7			13	16				
P8			12	5				
P9		18	14	8				
P10			13	17				
P11			12	16				
P12			11	5				
P13		6	13	8				
P14			12	17				
P15			11	16				
P16			10	5				
P17		7	12	8				
P18			11	17				
P19			10	16				
P20			2	5				
P21		9	11	8				
P22		9	10	17				

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TABLE XIII. Threshold voltage (gate to input) check test - Continued.

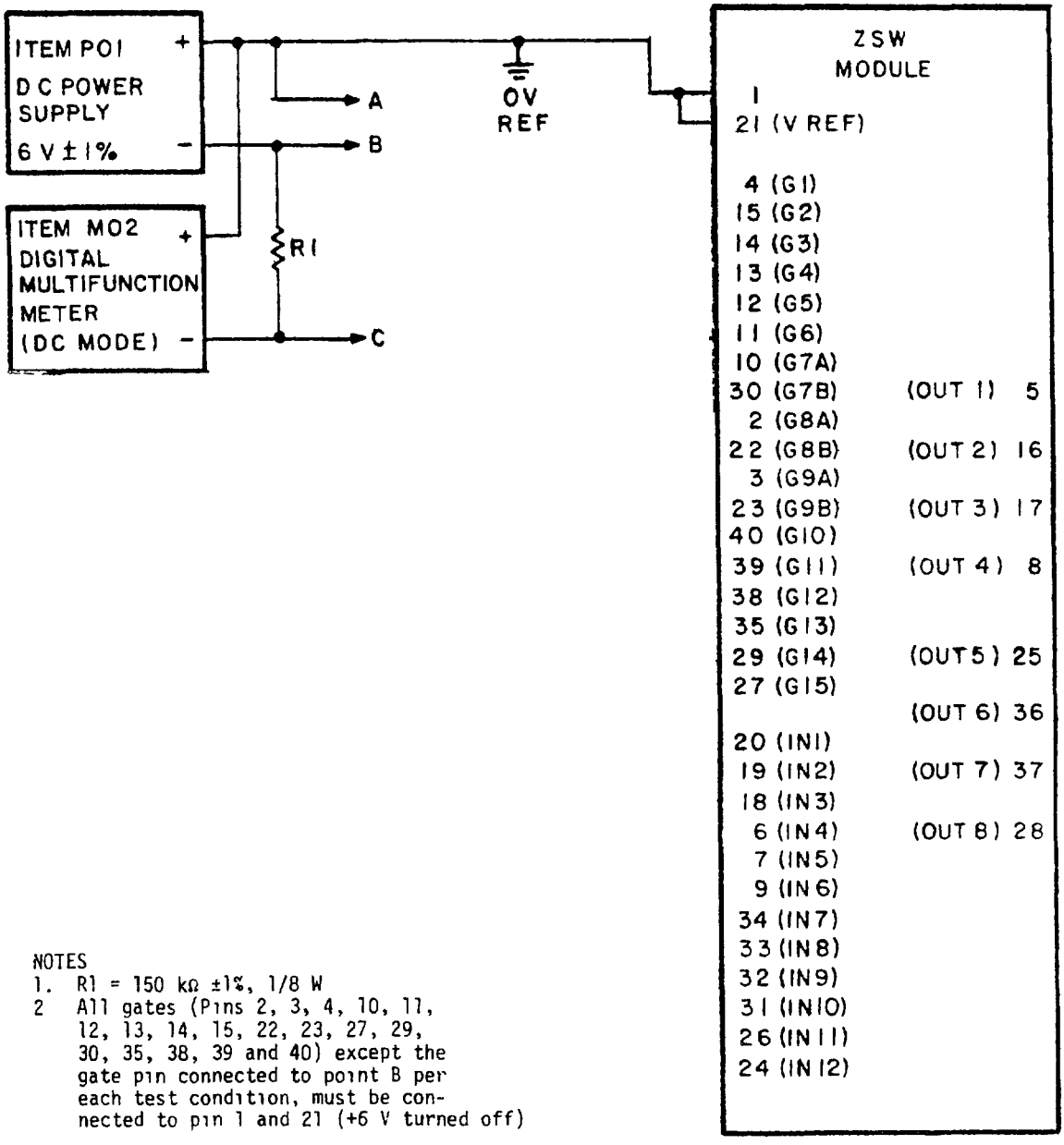
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P23	Threshold voltage, gate to input, check	9	2	16	0	200	mV	12
P24		9	3	5				
P25		34	30	28				
P26			22	37				
P27			23	36				
P28			40	25				
P29		33	22	28				
P30			23	37				
P31			40	36				
P32			39	25				
P33		32	23	28				
P34			40	37				
P35			39	36				
P36			38	25				
P37		31	40	28				
P38			39	37				
P39			38	36				
P40			35	25				
P41		26	39	28				
P42		26	38	37				
P43		26	35	36				

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TABLE XIII. Threshold voltage (gate to input) check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P44	Threshold voltage, gate to input, check	26	29	25	0	200	mV	12
P45	↓	24	38	28	↓	↓	↓	↓
P46		↓	35	37				
P47			29	36				
P48			27	25				

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NOTES
1. R1 = 150 kΩ ±1%, 1/8 W
2 All gates (Pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point B per each test condition, must be connected to pin 1 and 21 (+6 V turned off)

FIGURE 12 Threshold voltage (gate to input) check test circuit for table XIII

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TABLE XIV. Output isolation test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	Output isolation	13	14	20	5	Open	0.0	1.7	mVrms	13
Q2	↓	12	14	19	↓	↓	↓	↓	↓	↓
Q3		11	14	18						
Q4		10	40	6, 34						
Q5		2	23	7, 34						
Q6		3	22	9, 34						
Q7		13	30	20, 34						
Q8		14	13	20	16					
Q9		13	14	19		20				
Q10		12	14	18		20				
Q11		11	40	6, 34		Open				
Q12		10	23	7, 34						
Q13		2	22	9, 34						
Q14		14	30	20, 34						
Q15		15	13	20	17					
Q16		14	13	19		20				
Q17		13	14	18		19				
Q18		12	40	6, 34		Open				
Q19		11	23	7, 34						
Q20		10	22	9, 34						
Q21		15	30	20, 34						
Q22		4	13	20	8					

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TABLE XIV. Output isolation test - Continued.

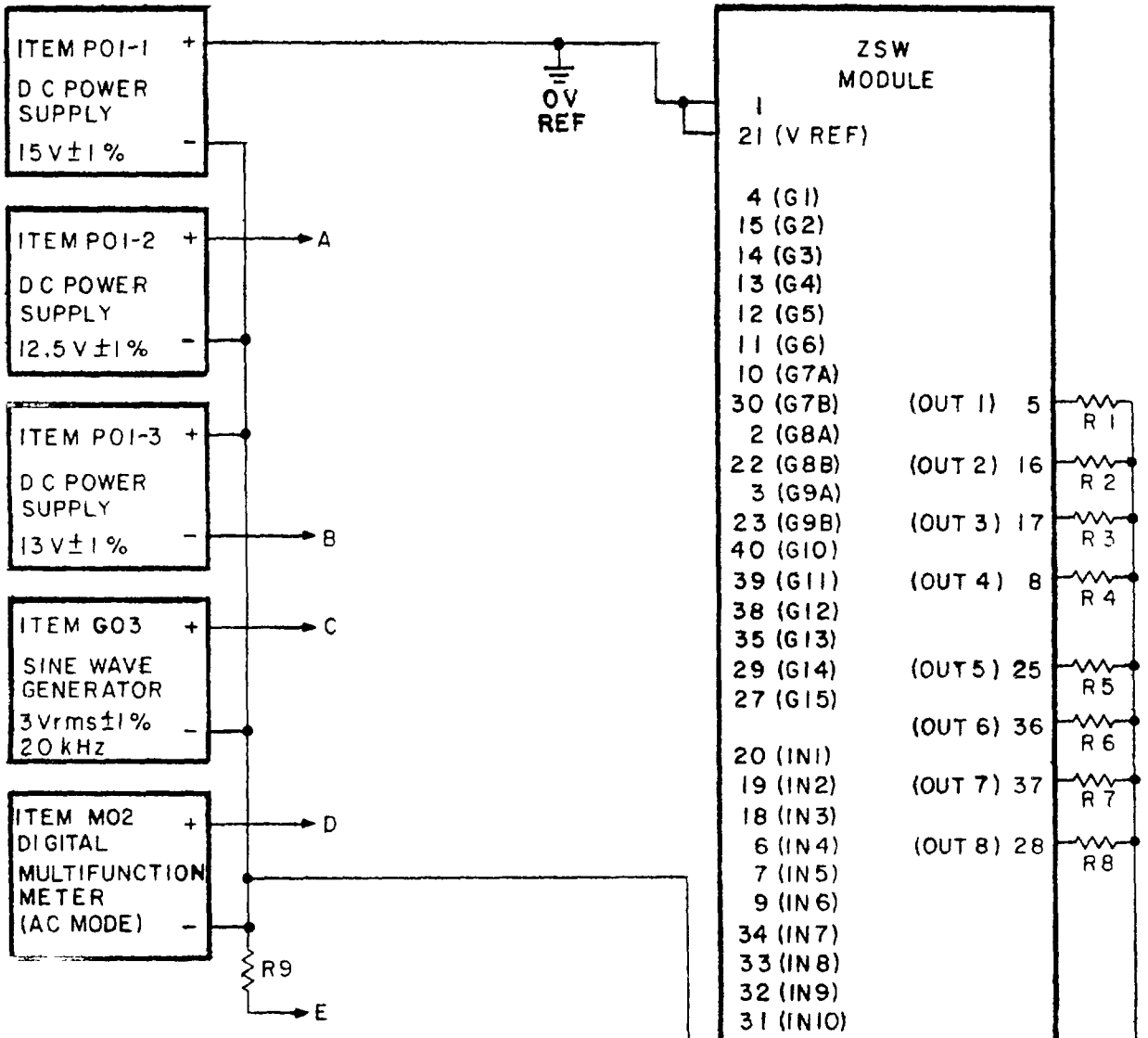
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q23	Output isolation	15	13	19	8	Open	0.0	1 7	mVrms	13
Q24		14	13	18						
Q25		13	40	6, 34						
Q26		12	23	7, 34						
Q27		11	22	9, 34						
Q28		4	30	20, 34						
Q29		40	23	34	25					
Q30		39	23	33						
Q31		38	23	32						
Q32		35	13	31, 20						
Q33		29	14	26, 20						
Q34		27	15	24, 20						
Q35		40	4	34, 20						
Q36		23	40	34	36					
Q37		40	23	33		34				
Q38		39	23	32		34				
Q39		38	13	31, 20		Open				
Q40		35	14	26, 20						
Q41		29	15	24, 20						
Q42		23	4	34, 20						
Q43		22	40	34	37					
Q44		23	40	33	37	34				

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TABLE XIV. Output isolation test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q45	Output isolation	40	23	32	37	33	0.0	1.7	mVrms	13
Q46	↓	39	13	31, 20	↓	Open	↓	↓	↓	↓
Q47		38	14	26, 20		↓				
Q48		35	15	24, 20						
Q49		22	4	34, 20						
Q50		30	40	34	28					
Q51		22	40	33	↓					
Q52		23	40	32						
Q53		40	13	31, 20						
Q54		39	14	26, 20						
Q55		38	15	24, 20						
Q56	↓	30	4	34, 20	↓	↓	↓	↓	↓	↓

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NOTES:
1. R1-R9 = 1 kΩ ±1%, 1/8 W
2. All gates (pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pins connected to point B per each test condition must be connected to point A

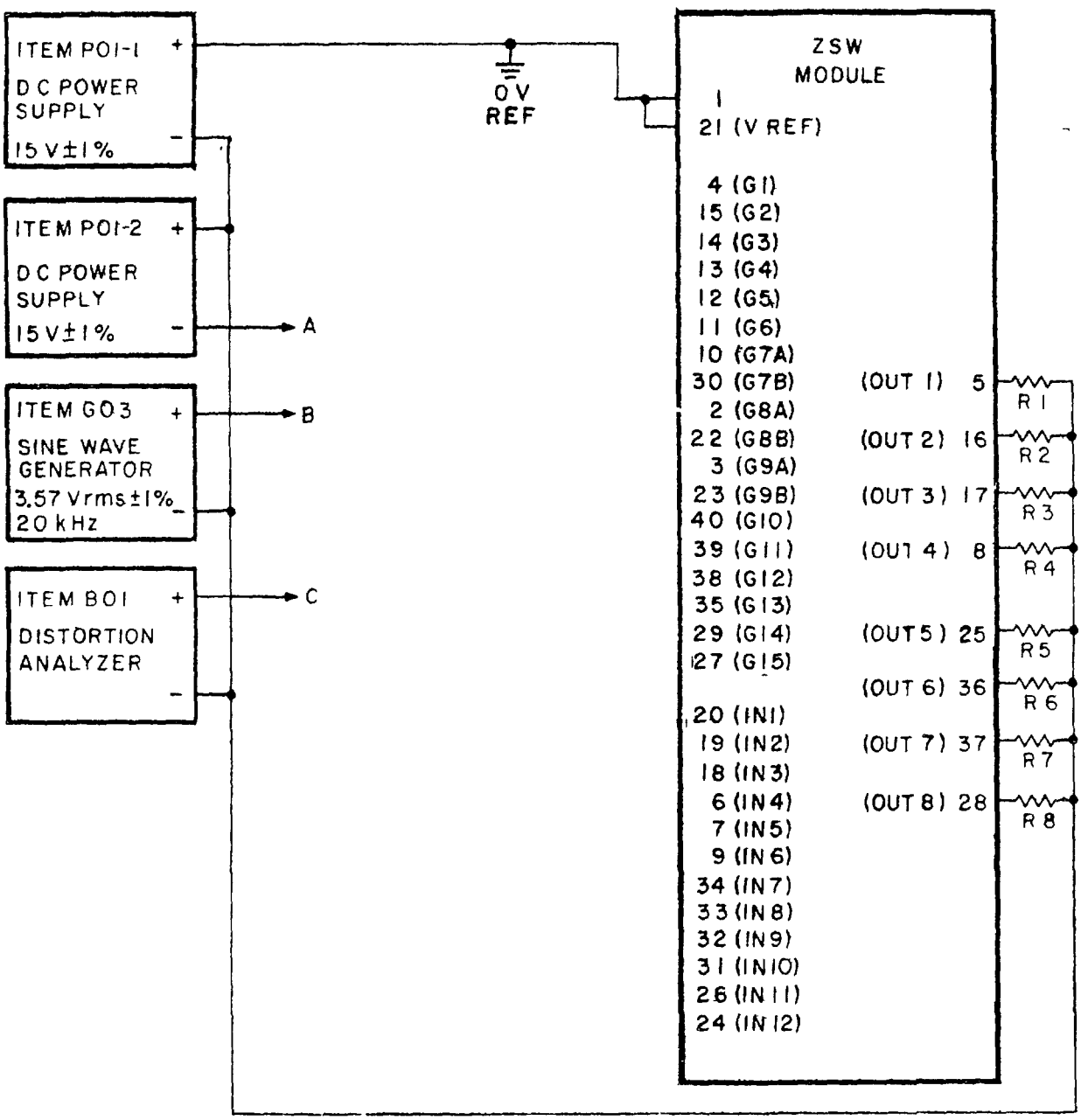
FIGURE 13 Output isolation test circuit for table XIV.

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TABLE XV. Output distortion test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Output distortion	13	20	5	0.0	0.5	%	14
R2	↓	↓	19	16	↓	↓	↓	↓
R3			18	17				
R4		↓	6	8				
R5		2	7	5				
R6		2	9	16				
R7		40	34	25				
R8		↓	33	36				
R9			32	37				
R10		↓	31	28				
R11		29	26	25				
R12	↓	29	24	36	↓	↓	↓	↓

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NOTES
1. R1-R8 = 20 kΩ ±1%, 1/8 W.
2 Measure distortion of item G03 and subtract this value from all output distortion measurements
3 All gates (Pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point A per each test condition, must be connected to pin 1 and 21 (+15 V, turned off)

FIGURE 14 Output distortion test circuit for table XV

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TABLE XVI. Output switching time (high-level, negative 5 V output) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S1	Output switching time (high level), negative 5 V output	Open	9	Open	5	3	0	6,000	ns	15, 16
S2			7		5	2				
S3			9		16	2				
S4			6		5	10				
S5			7		16	10				
S6			9		17	10				
S7			18		5	11				
S8			6		16					
S9			7		17					
S10			9		8					
S11			19		5	12				
S12			18		16					
S13			6		17					
S14			7		8					
S15			20		5	13				
S16			19		16					
S17			18		17					
S18			6		8					
S19			20		16	14				
S20			19		17	14				

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TABLE XVI. Output switching time (high-level, negative 5 V output) test - Continued

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S21	Output switching time (high level), negative 5 V output	Open	18	Open	8	14	0	6,000	ns	15, 16
S22			20		17	15				
S23			19		8	15				
S24			20		8	4				
S25			24		25	27				
S26			26		25	29				
S27			24		36	29				
S28			31		25	35				
S29			26		36	35				
S30			24		37	35				
S31			32		25	38				
S32			31		36					
S33			26		37					
S34			24		28					
S35			33		25	39				
S36			32		36	39				
S37			31		37	39				

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TABLE XVI. Output switching time (high-level, negative 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S38	Output switching time (high level) negative 5 V output	Open	26	Open	28	39	0	6,000	ns	15, 16
S39			34		25	40				
S40			33		36					
S41			32		37					
S42			31		28					
S43			34		36	23				
S44			33		37	23				
S45			32		28	23				
S46			34		37	22				
S47			33		28	22				
S48			34		28	30				

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TABLE XVII. Output switching time (low-level, negative 5 V output) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	Output switching time (low level), negative 5 V output	Open	9	Open	5	3	0	200	ns	15, 16
T2			7		5	2				
T3			9		16	2				
T4			6		5	10				
T5			7		16	10				
T6			9		17	10				
T7			18		5	11				
T8			6		16					
T9			7		17					
T10			9		8					
T11			19		5	12				
T12			18		16					
T13			6		17					
T14			7		8					
T15			20		5	13				
T16			19		16					
T17			18		17					
T18			6		8					
T19			20		16	14				
T20			19		17	14				

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TABLE XVII. Output switching time (low-level, negative 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T21	Output switching time (low level), negative 5 V output	Open	18	Open	8	14	0	200	ns	15, 16
T22			20		17	15				
T23			19		8	15				
T24			20		8	4				
T25			24		25	27				
T26			26		25	29				
T27			24		36	29				
T28			31		25	35				
T29			26		36	35				
T30			24		37	35				
T31			32		25	38				
T32			31		36					
T33			26		37					
T34			24		28					
T35			33		25	39				
T36			32		36	39				
T37			31		37	39				

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TABLE XVII. Output switching time (low-level, negative 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T38	Output switching time (low level) negative 5 V output	Open	26	Open	28	39	0	200	ns	15, 16
T39	↓	↓	34	↓	25	40	↓	↓	↓	↓
T40			33		36	↓				
T41			32		37					
T42			31		28					
T43			34		36	23				
T44			33		37	23				
T45			32		28	23				
T46			34		37	22				
T47			33		28	22				
T48			34		28	30				

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TABLE XVIII. Output switching time (high-level, 0 V output) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U1	Output switching time (high level), 0 V output	9	Open	Open	5	3	0	5,000	ns	15, 16
U2		7			5	2				
U3		9			16	2				
U4		6			5	10				
U5		7			16	10				
U6		9			17	10				
U7		18			5	11				
U8		6			16					
U9		7			17					
U10		9			8					
U11		19			5	12				
U12		18			16					
U13		6			17					
U14		7			8					
U15		20			5	13				
U16		19			16					
U17		18			17					
U18		6			8					
U19		20			16	14				
U20		19			17	14				

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TABLE XVIII. Output switching time (high-level, 0 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U21	Output switching time (high level), 0 V output	18	Open	Open	8	14	0	5,000	ns	15, 16
U22		20			17	15				
U23		19			8	15				
U24		20			8	4				
U25		24			25	27				
U26		26			25	29				
U27		24			36	29				
U28		31			25	35				
U29		26			36	35				
U30		24			37	35				
U31		32			25	38				
U32		31			36					
U33		26			37					
U34		24			28					
U35		33			25	39				
U36		32			36	39				
U37		31			37	39				

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TABLE XVIII. Output switching time (high-level, 0 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U38	Output switching time (high level) 0 V output	26	Open	Open	28	39	0	5,000	ns	15, 16
U39		34			25	40				
U40		33			36					
U41		32			37					
U42		31			28					
U43		34			36	23				
U44		33			37	23				
U45		32			28	23				
U46		34			37	22				
U47		33			28	22				
U48		34			28	30				

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TABLE XIX. Output switching time (low level, 0 V output) test.

Test condition	Parameter	Connect. A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	Output switching time (low level), 0 V output	9	Open	Open	5	3	0	200	ns	15, 16
V2		7			5	2				
V3		9			16	2				
V4		6			5	10				
V5		7			16	10				
V6		9			17	10				
V7		18			5	11				
V8		6			16					
V9		7			17					
V10		9			8					
V11		19			5	12				
V12		18			16					
V13		6			17					
V14		7			8					
V15		20			5	13				
V16		19			16					
V17		18			17					
V18		6			8					
V19		20			16	14				
V20		19			17	14				

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TABLE XIX. Output switching time (low level, 0 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V21	Output switching time (low level), 0 V output	18	Open	Open	8	14	0	200	ns	15, 16
V22		20			17	15				
V23		19			8	15				
V24		20			8	4				
V25		24			25	27				
V26		26			25	29				
V27		24			36	29				
V28		31			25	35				
V29		26			36	35				
V30		24			37	35				
V31		32			25	38				
V32		31			36	↓				
V33		26			37					
V34		24			28					
V35		33			25	39				
V36		32			36	39				
V37	↓	31	↓	↓	37	39	↓	↓	↓	↓

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TABLE XIX. Output switching time (low level, 0 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V38	Output switching time (low level) 0 V output	26	Open	Open	28	39	0	200	ns	15, 16
V39	↓	34	↓	↓	25	40	↓	↓	↓	↓
V40	↓	33	↓	↓	36	↓	↓	↓	↓	↓
V41	↓	32	↓	↓	37	↓	↓	↓	↓	↓
V42	↓	31	↓	↓	28	↓	↓	↓	↓	↓
V43	↓	34	↓	↓	36	23	↓	↓	↓	↓
V44	↓	33	↓	↓	37	23	↓	↓	↓	↓
V45	↓	32	↓	↓	28	23	↓	↓	↓	↓
V46	↓	34	↓	↓	37	22	↓	↓	↓	↓
V47	↓	33	↓	↓	28	22	↓	↓	↓	↓
V48	↓	34	↓	↓	28	30	↓	↓	↓	↓

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TABLE XX. Output switching time (high-level, positive 5 V output) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	Output switching time (high level), positive 5 V output	Open	Open	9	5	3	0	2,500	ns	15, 16
W2				7	5	2				
W3				9	16	2				
W4				6	5	10				
W5				7	16	10				
W6				9	17	10				
W7				18	5	11				
W8				6	16					
W9				7	17					
W10				9	8					
W11				19	5	12				
W12				18	16					
W13				6	17					
W14				7	8					
W15				20	5	13				
W16				19	16					
W17				18	17					
W18				6	8					
W19				20	16	14				
W20				19	17	14				

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TABLE XX. Output switching time (high-level, positive 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W21	Output switching time (high level), positive 5 V output	Open	Open	18	8	14	0	2,500	ns	15, 16
W22				20	17	15				
W23				19	8	15				
W24				20	8	4				
W25				24	25	27				
W26				26	25	29				
W27				24	36	29				
W28				31	25	35				
W29				26	36	35				
W30				24	37	35				
W31				32	25	38				
W32				31	36					
W33				26	37					
W34				24	28					
W35				33	25	39				
W36				32	36	39				
W37				31	37	39				

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TABLE XX. Output switching time (high-level, positive 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W38	Output switching time (high level) positive 5 V output	Open	Open	26	28	39	0	2,500	ns	15, 16
W39				34	25	40				
W40				33	36					
W41				32	37					
W42				31	28					
W43				34	36	23				
W44				33	37	23				
W45				32	28	23				
W46				34	37	22				
W47				33	28	22				
W48				34	28	30				

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TABLE XXI. Output switching time (low-level, positive 5 V output) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X1	Output switching time (low level), positive 5 V output	Open	Open	9	5	3	0	200	ns	15, 16
X2				7	5	2				
X3				9	16	2				
X4				6	5	10				
X5				7	16	10				
X6				9	17	10				
X7				18	5	11				
X8				6	16					
X9				7	17					
X10				9	8					
X11				19	5	12				
X12				18	16					
X13				6	17					
X14				7	8					
X15				20	5	13				
X16				19	16					
X17				18	17					
X18				6	8					
X19				20	16	14				
X20				19	17	14				

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TABLE XXI. Output switching time (low-level, positive 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X21	Output switching time (low level), positive 5 V output	Open	Open	18	8	14	0	200	ns	15, 16
X22				20	17	15				
X23				19	8	15				
X24				20	8	4				
X25				24	25	27				
X26				26	25	29				
X27				24	36	29				
X28				31	25	35				
X29				26	36	35				
X30				24	37	35				
X31				32	25	38				
X32				31	36					
X33				26	37					
X34				24	28					
X35				33	25	39				
X36				32	36	39				
X37				31	37	39				

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TABLE XXI. Output switching time (low-level, positive 5 V output) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X38	Output switching time (low level) positive 5 V output	Open	Open	26	28	39	0	200	ns	15, 16
X39				34	25	40				
X40				33	36					
X41				32	37					
X42				31	28					
X43				34	36	23				
X44				33	37	23				
X45				32	28	23				
X46				34	37	22				
X47				33	28	22				
X48				34	28	30				

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TABLE XXII. Output switching time (most positive peak, 0 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y1	Output switching time, most positive peak, 0 V output	9	Open	Open	5	3	0.0	2.5	V	15, 16
Y2		7			5	2				
Y3		9			16	2				
Y4		6			5	10				
Y5		7			16	10				
Y6		9			17	10				
Y7		18			5	11				
Y8		6			16					
Y9		7			17					
Y10		9			8					
Y11		19			5	12				
Y12		18			16					
Y13		6			17					
Y14		7			8					
Y15		20			5	13				
Y16		19			16					
Y17		18			17					
Y18		6			8					
Y19		20			16	14				
Y20		19			17	14				

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TABLE XXII. Output switching time (most positive peak, 0 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y21	Output switching time, most positive peak, 0 V output	18	Open	Open	8	14	0.0	2.5	V	15, 16
Y22	↓	20	↓	↓	17	15	↓	↓	↓	↓
Y23		19			8	15				
Y24		20			8	4				
Y25		24			25	27				
Y26		26			25	29				
Y27		24			36	29				
Y28		31			25	35				
Y29		26			36	35				
Y30		24			37	35				
Y31		32			25	38				
Y32		31			36	↓				
Y33		26			37					
Y34		24			28					
Y35		33			25	39				
Y36		32			36	39				
Y37	↓	31	↓	↓	37	39	↓	↓	↓	↓

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TABLE XXII Output switching time (most positive peak, 0 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y38	Output switching time, most positive peak, 0 V output	26	Open	Open	28	39	0.0	2.5	V	15, 16
Y39	↓	34	↓	↓	25	40	↓	↓	↓	↓
Y40		33			36	↓				
Y41		32			37					
Y42		31			28					
Y43		34			36	23				
Y44		33			37	23				
Y45		32			28	23				
Y46		34			37	22				
Y47		33			28	22				
Y48	↓	34	↓	↓	28	30	↓	↓	↓	↓

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TABLE XXIII. Output switching time (most negative peak, 0 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	Output switching time, most negative peak, 0 V output	9	Open	Open	5	3	0.0	2.5	V	15, 16
Z2		7			5	2				
Z3		9			16	2				
Z4		6			5	10				
Z5		7			16	10				
Z6		9			17	10				
Z7		18			5	11				
Z8		6			16					
Z9		7			17					
Z10		9			8					
Z11		19			5	12				
Z12		18			16					
Z13		6			17					
Z14		7			8					
Z15		20			5	13				
Z16		19			16					
Z17		18			17					
Z18		6			8					
Z19		20			16	14				
Z20		19			17	14				

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TABLE XXIII. Output switching time (most negative peak, 0 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z21	Output switching time, most negative peak, 0 V output	18	Open	Open	8	14	0.0	2.5	V	15, 16
Z22	↓	20	↓	↓	17	15	↓	↓	↓	↓
Z23		19			8	15				
Z24		20			8	4				
Z25		24			25	27				
Z26		26			25	29				
Z27		24			36	29				
Z28		31			25	35				
Z29		26			36	35				
Z30		24			37	35				
Z31		32			25	38				
Z32		31			36	↓				
Z33		26			37					
Z34		24			28					
Z35		33			25	39				
Z36		32			36	39				
Z37	↓	31	↓	↓	37	39	↓	↓	↓	↓

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TABLE XXXII. Output switching time (most negative peak, 0 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z38	Output switching time, most negative peak, 0 V output	26	Open	Open	28	39	0.0	2.5	V	15, 16
Z39	↓	34	↓	↓	25	40	↓	↓	↓	↓
Z40	↓	33	↓	↓	36	↓	↓	↓	↓	↓
Z41	↓	32	↓	↓	37	↓	↓	↓	↓	↓
Z42	↓	31	↓	↓	28	↓	↓	↓	↓	↓
Z43	↓	34	↓	↓	36	23	↓	↓	↓	↓
Z44	↓	33	↓	↓	37	23	↓	↓	↓	↓
Z45	↓	32	↓	↓	28	23	↓	↓	↓	↓
Z46	↓	34	↓	↓	37	22	↓	↓	↓	↓
Z47	↓	33	↓	↓	28	22	↓	↓	↓	↓
Z48	↓	34	↓	↓	28	30	↓	↓	↓	↓

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TABLE XXIV. Output switching time (most positive peak, +5 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	Output switching time time, most positive peak, +5 V output	Open	Open	9	5	3	0.0	2.5	V	15, 16
AA2				7	5	2				
AA3				9	16	2				
AA4				6	5	10				
AA5				7	16	10				
AA6				9	17	10				
AA7				18	5	11				
AA8				6	16					
AA9				7	17					
AA10				9	8					
AA11				19	5	12				
AA12				18	16					
AA13				6	17					
AA14				7	8					
AA15				20	5	13				
AA16				19	16					
AA17				18	17					
AA18				6	8					
AA19				20	16	14				
AA20				19	17	14				

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ABLE XXIV. Output switching time (most positive peak, +5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA21	Output switching time, most positive peak, +5 V output	Open	Open	18	8	14	0.0	2.5	V	15, 16
AA22				20	17	15				
AA23				19	8	15				
AA24				20	8	4				
AA25				24	25	27				
AA26				26	25	29				
AA27				24	36	29				
AA28				31	25	35				
AA29				26	36	35				
AA30				24	37	35				
AA31				32	25	38				
AA32				31	36					
AA33				26	37					
AA34				24	28					
AA35				33	25	39				
AA36				32	36	39				
AA37				31	37	39				

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TABLE XXIV. Output switching time (most positive peak, +5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA38	Output switching time, most positive peak, +5 V output	Open	Open	26	28	39	0.0	2.5	V	15, 16
AA39	↓	↓	↓	34	25	40	↓	↓	↓	↓
AA40				33	36	↓				
AA41				32	37					
AA42				31	28					
AA43				34	36	23				
AA44				33	37	23				
AA45				32	28	23				
AA46				34	37	22				
AA47				33	28	22				
AA48	↓	↓	↓	34	28	30	↓	↓	↓	↓

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TABLE XXV Output switching time (most negative peak, +5 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	Output switching time, most negative peak, +5 V output	Open	Open	9	5	3	0 0	2.5	V	15, 16
BB2				7	5	2				
BB3				9	16	2				
BB4				6	5	10				
BB5				7	16	10				
BB6				9	17	10				
BB7				18	5	11				
BB8				6	16					
BB9				7	17					
BB10				9	8					
BB11				19	5	12				
BB12				18	16					
BB13				6	17					
BB14				7	8					
BB15				20	5	13				
BB16				19	16					
BB17				18	17					
BB18				6	8					
BB19				20	16	14				
BB20				19	17	14				

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TABLE XXV Output switching time (most negative peak, +5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB21	Output switching time, most negative peak, +5 V output	Open	Open	18	8	14	0.0	2.5	V	15, 16
BB22				20	17	15				
BB23				19	8	15				
BB24				20	8	4				
BB25				24	25	27				
BB26				26	25	29				
BB27				24	36	29				
BB28				31	25	35				
BB29				26	36	35				
BB30				24	37	35				
BB31				32	25	38				
BB32				31	36					
BB33				26	37					
BB34				24	28					
BB35				33	25	39				
BB36				32	36	39				
BB37				31	37	39				

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TABLE XXV. Output switching time (most negative peak, +5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB38	Output switching time, most negative peak, +5 V output	Open	Open	26	28	39	0.0	2.5	V	15, 16
BB39				34	25	40				
BB40				33	36					
BB41				32	37					
BB42				31	28					
BB43				34	36	23				
BB44				33	37	23				
BB45				32	28	23				
BB46				34	37	22				
BB47				33	28	22				
BB48				34	28	30				

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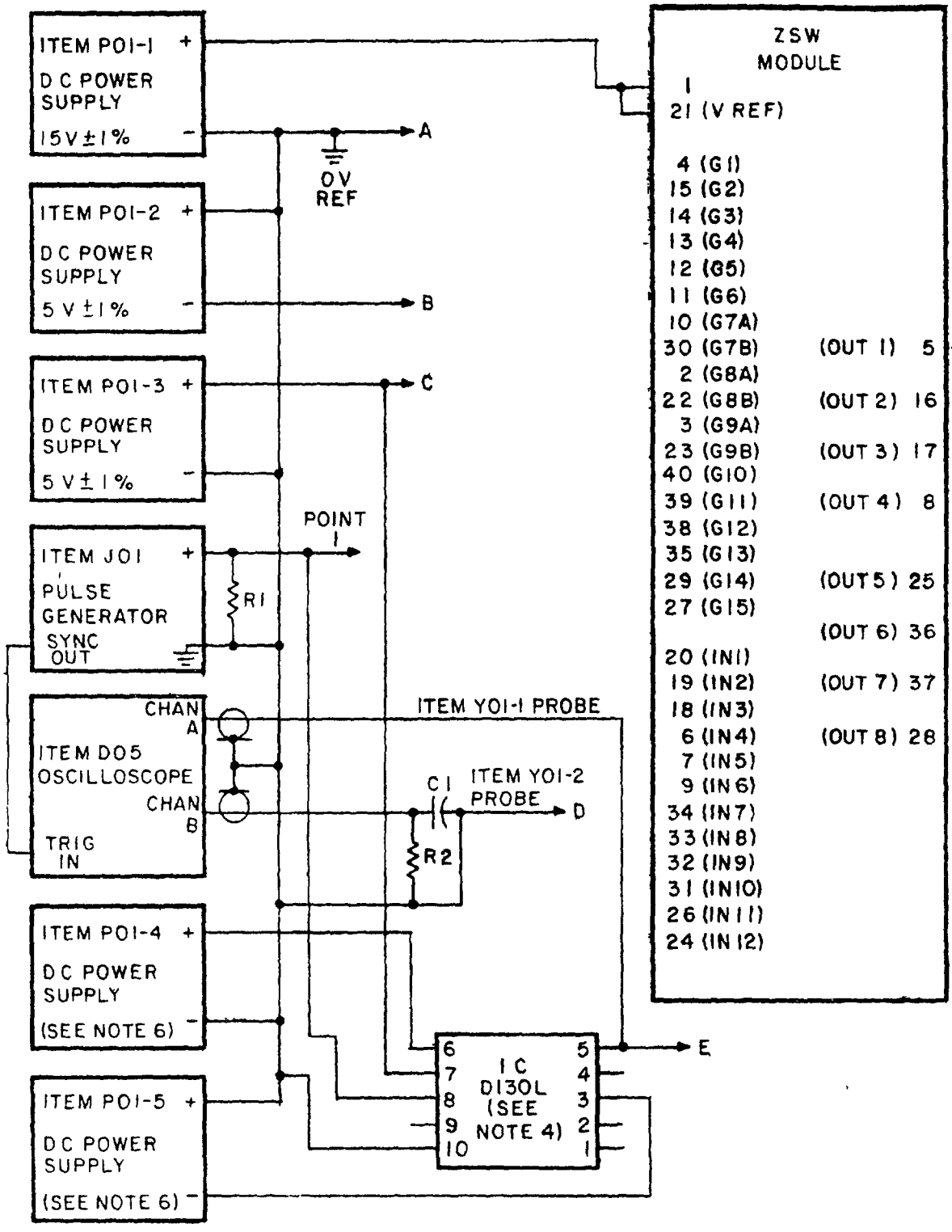
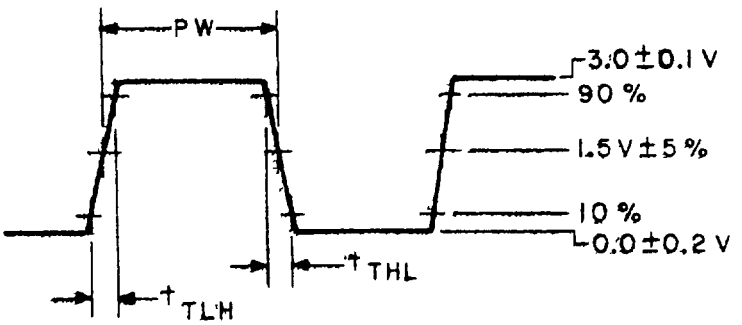


FIGURE 15. Output switching time (high-level, low-level, most positive peak and most negative peak) test circuit for tables XVI through XXV.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 20 kΩ ±1%, 1/8 W.
3. C1 = 30 pF ±5% (including probe and parasitic capacitance).
4. I.C. = Siliconix DI30L or equivalent.
5. Input pulse at point 1.



Input pulse parameters

PW = See note 6.
t_{TLH}, t_{THL} = 25 ±5 ns.

6. Initially adjust item P01-4 to 15 V and item P01-5 to 17 V, adjust item P01-4 to obtain a 13 ±0.2 V level and item P01-5 to obtain a -15 ±0.2 V level as specified in input pulse at point E on figure 16.
7. Adjust the pulse width and repetition rate of the waveform at point 1 to obtain a pulse width of 10 ±1 µs and a period of 20 ±2 µs for the input pulse at point E (figure 16).
8. All gates (Pins 2, 3, 4, 10, 11, 12, 13, 14, 15, 22, 23, 27, 29, 30, 35, 38, 39 and 40) except the gate pin connected to point E per each test condition, must be connected to pin 1 and 21 (+15 V, turned off)

FIGURE 15 Output switching time (high-level, low-level, most positive peak and most negative peak) test circuit for tables XVI through XXV - Continued.

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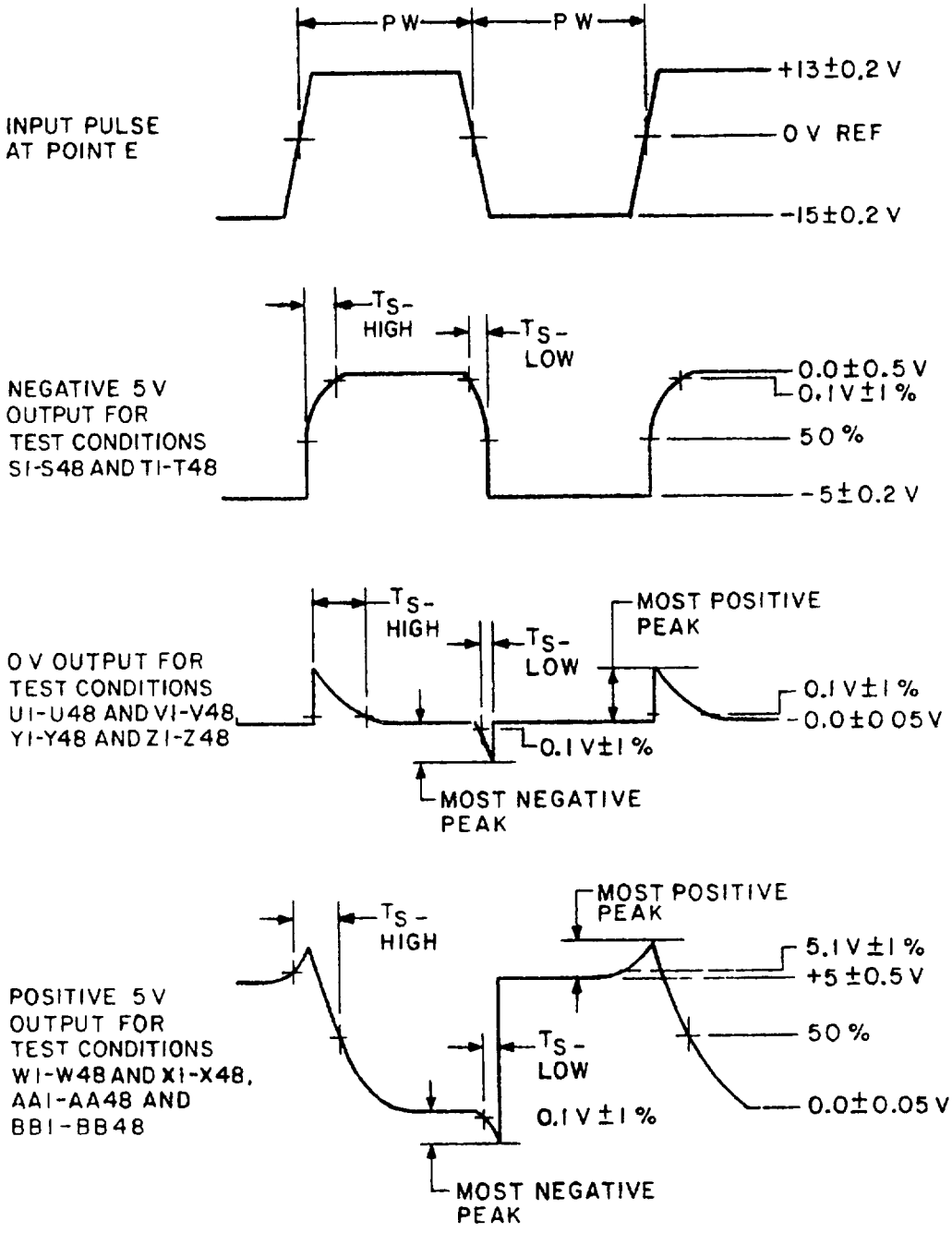


FIGURE 16 Output switching time (high-level, low-level, most positive peak and most negative peak) test circuit waveforms for tables XVI through XXV

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NOTES:
1 Test conditions S1-S48 and T1-T48:
  (a) Reference the negative 5 V output waveform.
  (b) T_{S-HIGH} level is measured from the 50% point of the positive edge to the point 0.1 V below 0 V on the positive edge.
  (c) T_{S-LOW} level is measured from the point 0.1 V below 0 V on the negative edge to the 50% point of the negative edge.
2 Test conditions U1-U48 and V1-V48:
  (a) Reference the 0 V output waveform
  (b) T_{S-HIGH} level is measured from the point 0.1 V above 0 V on the positive edge to the point 0.1 V above 0 V on the negative edge.
  (c) T_{S-LOW} level is measured from the point 0.1 V below 0 V on the negative edge to the point 0 1 V below 0 V on the positive edge
3 Test conditions W1-W48 and X1-X48:
  (a) Reference the positive 5 V output waveform.
  (b) T_{S-HIGH} level is measured from the point 5.1 V above 0 V on the positive peak to the 50% point of the negative edge.
  (c) T_{S-LOW} level is measured from the point 0.1 V below 0 V on the negative peak to the 50% point of the positive edge.
4. Test conditions Y1-Y48 and Z1-Z48
  (a) Reference the 0 V output waveform
  (b) Positive (+) peak is measured from the 0 V level to the most positive peak.
  (c) Negative (-) peak is measured from the 0 V level to the most negative peak.
5 Test conditions AA1-AA48 and BB1-BB48
  (a) Reference the positive 5 V output waveform
  (b) Positive (+) peak is measured from the 5 V level to the most positive peak
  (c) Negative (-) peak is measured from the 0 V level to the most negative peak.

FIGURE 16 Output switching time (high-level, low-level, most positive peak and most negative peak) test circuit waveforms for tables XVI through XXV - Continued.

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